Haoran Xu

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Education

University of Michigan Ann Arbor, Michigan
Ph.D. Student, Electrical and Computer Engineering Sep. 2026 - Present

Tsinghua University Beijing, China
Bachelor of Science in Electrical Engineering Sep. 2022 - Jul. 2026

Research Interests

- **Energy Storage Systems**: Physics-based modeling, state estimation (SOC/SOH), and optimal control of batteries.
- **Sustainable Transportation**: Developing advanced BMS and thermal management systems for electrified vehicles.
- **Grid Integration:** Leveraging energy storage to bridge the gap between transportation sectors and renewable power grids.

Honors & Awards

- Excellent Graduate, Tsinghua University Jun. 2026

Publications & Manuscripts

- [**ICML 2026**] Fang, Y., **Xu, H.**, Han, J., Ding, S., Wang, Y., Wang, Y., & Wang, X. (2025). "BioArc: Discovering Optimal Neural Architectures for Biological Foundation Models."

Research Experiences & Selected Projects

Knowledge-Augmented LLM Agent for Concurrency Bug Localization Jun. 2025 - Present
Advisor: Tianyi Zhang & Yongle Zhang, Assistant Professors, Purdue University

- Developing an LLM-based agent for static root-cause localization of concurrency bugs in distributed systems
- Curating and analyzing real-world concurrency bugs from Apache projects including Kafka, Spark, HDFS, HBase, and ZooKeeper
- Building a knowledge-augmented debugging framework that integrates code facts and diagnostic knowledge to guide bug localization

Neural Architecture Search for Single-Cell Foundation Models Jul. 2025 - Jan. 2026
Advisor: Xuan Wang, Assistant Professor, Virginia Tech

- Investigating the optimal neural architectures for building foundation models on single-cell sequencing data
- Utilizing NNI with PyTorch Lightning to explore CNN, LSTM, Mamba, Hyena, and Transformer across multiple tasks
- Developed the "BioArc Agent," a multi-agent LLM that consistently outperforms standard predictors in recommending optimal architectures for novel tasks

Skills

- **Programming Language:** C, C++, Python, Java, Fortran, Shell
- **Software & Tools:** Mathematica, Matlab, Vivado, LTSpice, PyTorch Lightning

Leadership & Activities

- Vice Minister, Academic Affairs Department, Tsinghua University Student Union
- Vice Minister, Technical Department, Innovation Association of the Department of Electrical Engineering, Tsinghua University
- Five-Star Tsinghua Volunteer (500+ accumulated service hours)